

Yinda Zhang

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RESEARCH INTERESTS Systems and Networking, Streaming Algorithms

EDUCATION **M.S., University of Chicago** Sep. 2020 – Dec. 2021
Advisor: Prof. [Junchen Jiang](#)
B.S., Peking University Sep. 2016 – Jul. 2020
Advisor: Prof. [Tong Yang](#)

RESEARCH EXPERIENCE **Research Assistant** Jul 2020 – Present
Advisor: Prof. [Junchen Jiang](#)

Measurement study over users' quality sensitivity

- Analyzed the interplay between video content and quality sensitivity in the wild using real-world measurements from 7.6M video sessions in four popular content providers

Applying influence function to machine learning algorithms

- Used the influence function to retrain machine learning algorithms

Research Assistant Jul 2020 – Present
Advisors: Prof. [Zaoxing Liu](#) and Prof. [Junchen Jiang](#)

Applying sketching algorithms to network measurement systems

- Defined arbitrary partial key query and designed a sketch that can efficiently address the problem in both software and hardware platforms
- Designed a framework that can be applied to many sketches and enable them to work on multi-core architecture

Research Assistant May 2018 – Present
Advisor: Prof. [Tong Yang](#)

Designing approximate streaming algorithms for different applications

- Designed an adaptive communication scheme for geo-distributed systems (*e.g.*, federated learning) which need to aggregate the data of multiple local devices over wide area networks
- Defined periodicity and designed a sketch to find periodic items in data streams
- Designed a sketch that can achieve high accuracy and high throughput for both persistence estimation and finding persistent items
- Designed a framework that can be applied to many sketches and enable them to support queries in sliding windows
- Designed a framework that can adapt many scheduling problems for a limited number of queues and work on Tofino switches

PUBLICATIONS

Conference Papers

1. Yikai Zhao*, **Yinda Zhang***, Yuanpeng Li* (* equal contribution), Yi Zhou, Chen Chunhui, Tong Yang, Bin Cui, “MinMax Sampling: A Near-optimal Global Summary for Aggregation in the Wide Area”, *To appear in SIGMOD 2022*
2. Zhuochen Fan, **Yinda Zhang**, Tong Yang, Mingyi Yan, Gang Wen, Yuhan Wu, Hongze Li, Bin Cui, “PeriodicSketch: Finding Periodic Items in Data Streams”, *To appear in ICDE 2022*
3. **Yinda Zhang**, Zaoxing Liu, Ruixin Wang, Tong Yang, Jizhou Li, Ruijie Miao, Peng Liu, Ruwen Zhang, Junchen Jiang, “CocoSketch: High-Performance Sketch-based Measurement over Arbitrary Partial Key Query”, *SIGCOMM 2021*
4. **Yinda Zhang**, Jinyang Li, Yutian Lei, Tong Yang, Zhetao Li, Gong Zhang, Bin Cui, “On-Off Sketch: A Fast and Accurate Sketch on Persistence”, *VLDB 2021*
5. Xiangyang Gou*, Long He*, **Yinda Zhang***, Ke Wang, Xilai Liu, Tong Yang, Yi Wang, Bin Cui, “Sliding Sketches: A Framework using Time Zones for Data Stream Processing in Sliding Windows”, *KDD 2020*
6. Tong Yang, Jizhou Li, Yikai Zhao, Kaicheng Yang, Hao Wang, Jie Jiang, **Yinda Zhang**, Nicholas Zhang, “QCluster: Clustering Packets for Flow Scheduling”, *Under Review*

WORK

EXPERIENCE

Research Intern, Conviva

Jul. 2020 – Jan. 2021